

Texsolv AB  
Vitlanda Gård  
662 98 Tösse

## Tensile strength test

(1 appendix)

### 1 Introduction

By commission of Texsolv AB tensile tests were performed on ropes.

Test place: SP Structural and Solid Mechanics in Borås.

### 2 Test objects

Designation: Ropes with a length of 2 m:s marked No 4, No 6, No 8, No 12, No 16 and No 20, see photos 1-2.



Photo 1. Test objects.



Photo 2. Test objects.

Selection of test objects: The test objects have been selected by the client without SP's assistance.

Arrival of test objects: 2016-01-18.

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#### SP Technical Research Institute of Sweden

Postal address  
SP  
Box 857  
SE-501 15 BORÅS  
Sweden

Office location  
Västeråsen  
Brinellgatan 4  
SE-504 62 BORÅS

Phone / Fax / E-mail  
+46 10 516 50 00  
+46 33 13 55 02  
info@sp.se

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### 3 Test method and implement

Test method: The test was performed in a tensile testing machine. As attachment to the ropes, shackles were used with diameter 16.4 mm, test setup see photo 3. The total length (including loops) at pre-load 0.1 kN was measured for the test objects in tensile strength test, see table 1. Test objects were then subjected to an increasing tensile load until fracture occurred with a constant speed of 50 mm/min. The elongations are measured with the positioning of the cylinder and some deviations due to machine stiffness and settings in the fixtures can occur.



**Photo 3. Test setup.**

Number of tests: 6  
Test date: 2016-01-27  
Measurements: Breaking force and elongation was registered

### 4 Results

The test results are shown in table 1. The test results shown in this report refer only to the tested objects.

**Table 1. Results from tensile strength test.**

| Test marking | Breaking-force kN | Breaking-force kg | Total length mm | Elongation % | Remarks                           |
|--------------|-------------------|-------------------|-----------------|--------------|-----------------------------------|
| No 4         | 2.05              | 209               | 1068            | 20.8         | Fracture at the loop, see photo 4 |
| No 6         | 2.93              | 299               | 1027            | 21.3         | Fracture at the loop, see photo 5 |
| No 8         | 3.44              | 351               | 1057            | 22.3         | Fracture at the loop, see photo 6 |
| No 12        | 4.80              | 489               | 1061            | 22.1         | Fracture at the loop, see photo 7 |
| No 16        | 6.30              | 642               | 1046            | 22.5         | Fracture at the loop, see photo 8 |
| No 20        | 8.27              | 843               | 1013            | 25.0         | Fracture at the loop, see photo 9 |

Breaking force converted to kg has  $g = 9.81$  been used.



Photo 4. Fracture, test object marked No 4.



Photo 5. Fracture, test object marked No 6.



Photo 6. Fracture, test object marked No 8.



Photo 7. Fracture, test object marked No 12.



Photo 8 . Fracture, test object marked No 16.



Photo 9. Fracture, test object marked No 20.

## 5 Measurement uncertainty

The total calculated measurement uncertainty for the breaking force is  $< 1\%$ . Reported uncertainty corresponds to an approximate 95 % confidence interval around the measured value. The interval has been calculated in accordance with EA-4/16 (EA guidelines on the expression of uncertainty in quantitative testing), which is normally accomplished by quadratic addition of the actual standard uncertainties and multiplication of the resulting combined standard uncertainty by the coverage factor  $k=2$ .

### SP Technical Research Institute of Sweden SP Structural and Solid Mechanics - Life and Reliability

Performed by

*Jukka Holappa*

Jukka Holappa

Signed by: Jukka Holappa  
Reason: I am the author of this document  
Date & Time: 2016-02-04 08:22:44 +01:00

Examined by

*Billy Alvarsson*

Billy Alvarsson

Signed by: Billy Alvarsson  
Reason: I have reviewed the document  
Date & Time: 2016-02-04 09:59:46 +01:00

## Appendix

### 1. Diagrams

## Appendix 1

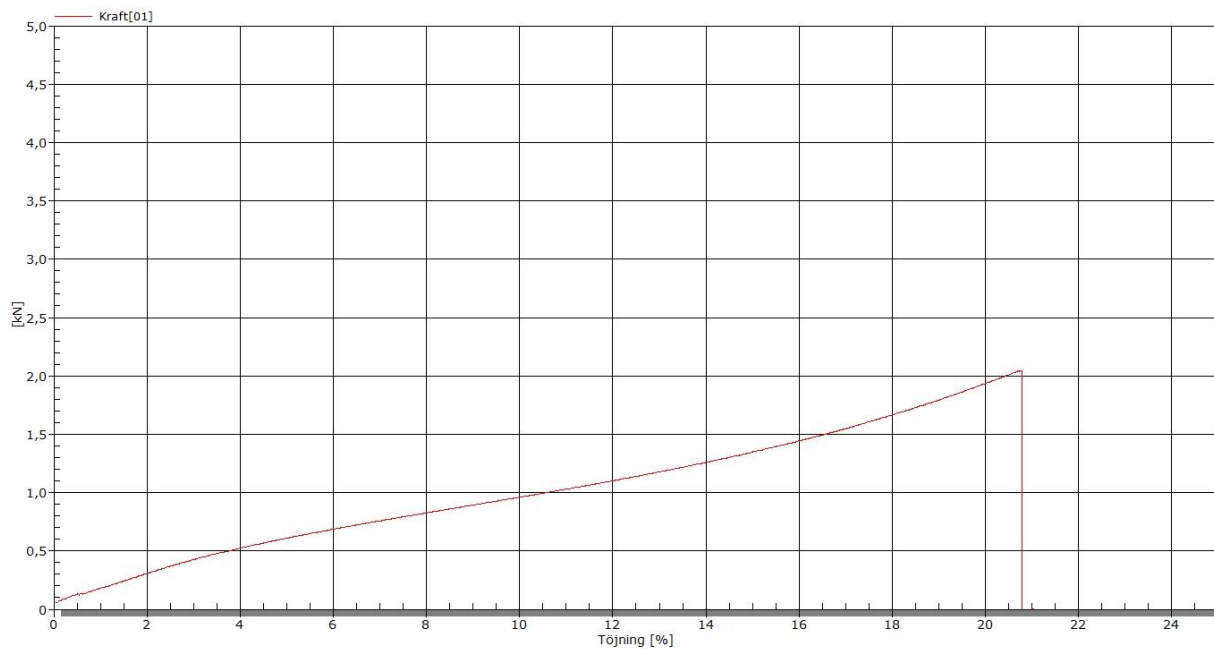


Diagram 1. Test marking No 4.

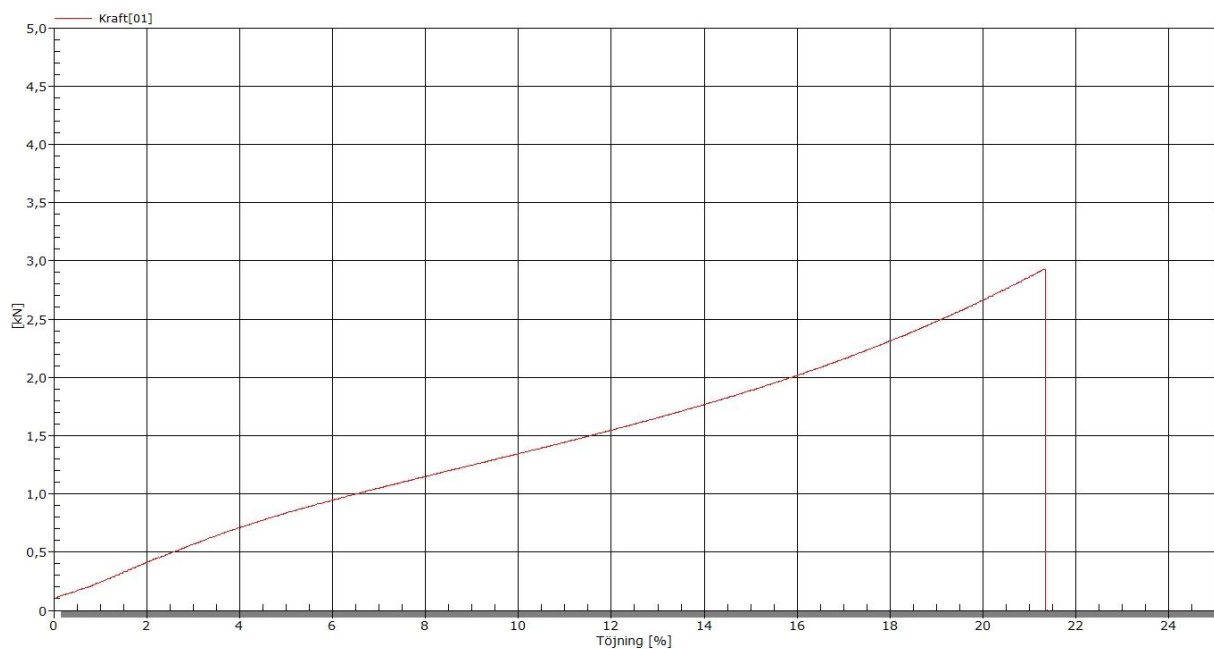


Diagram 2. Test marking No 6.

## Appendix 1

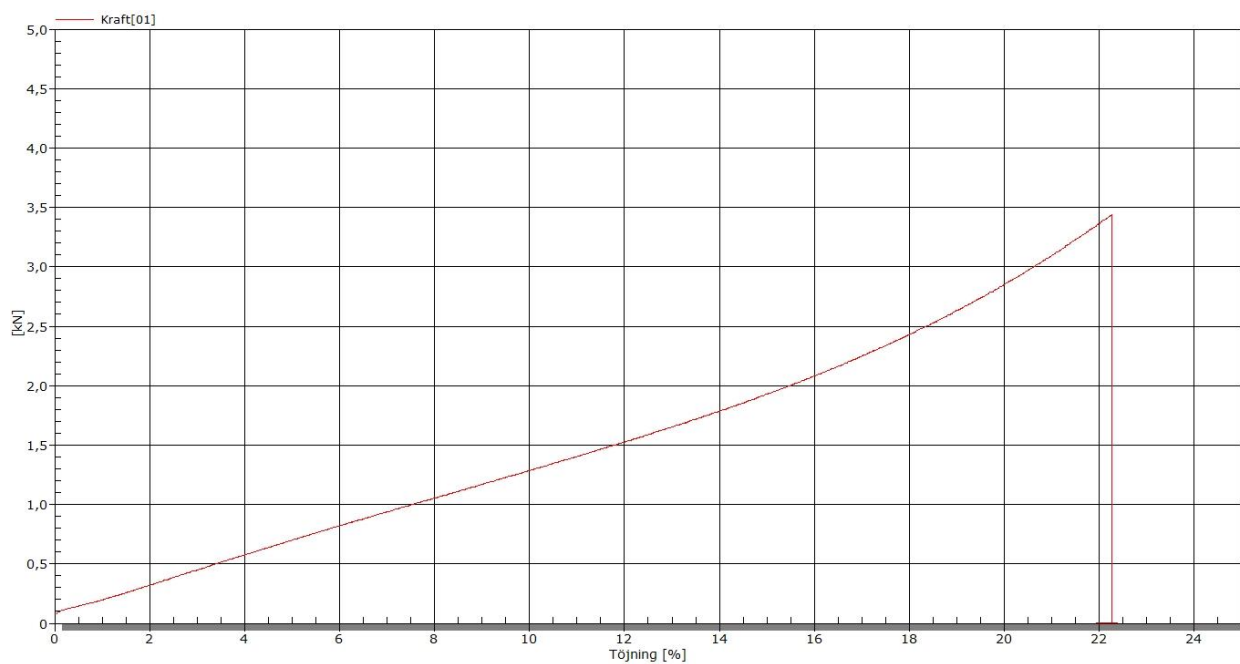


Diagram 3. Test marking No 8.

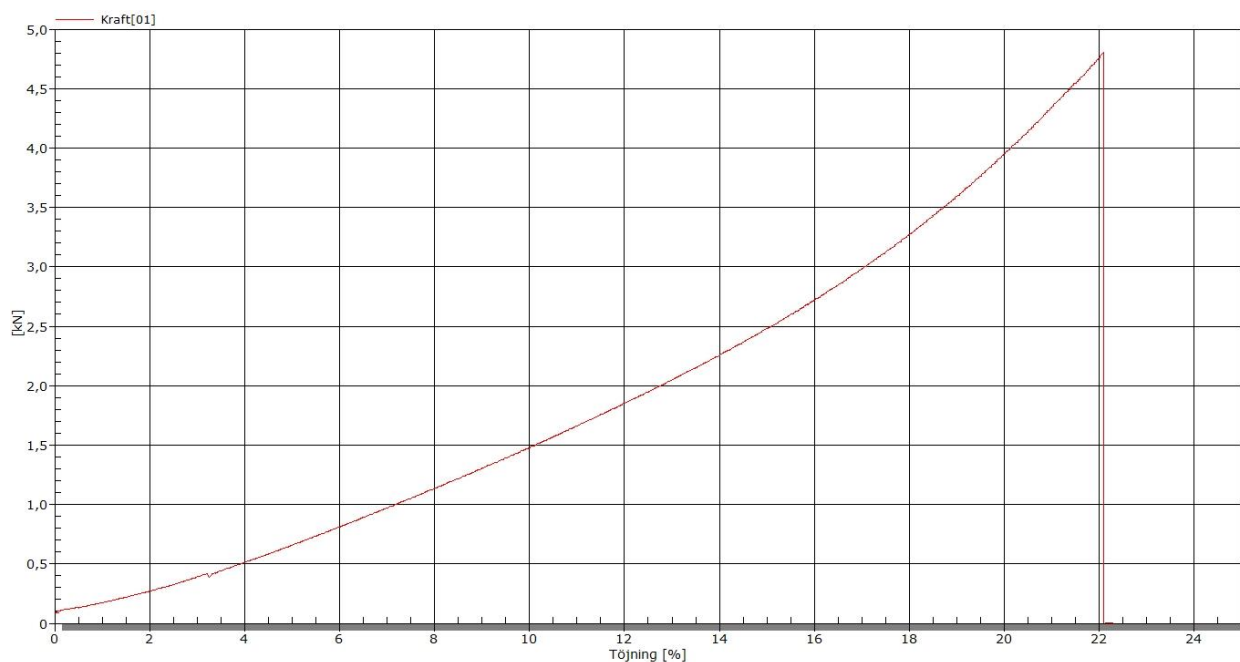


Diagram 4. Test marking No 12.

## Appendix 1

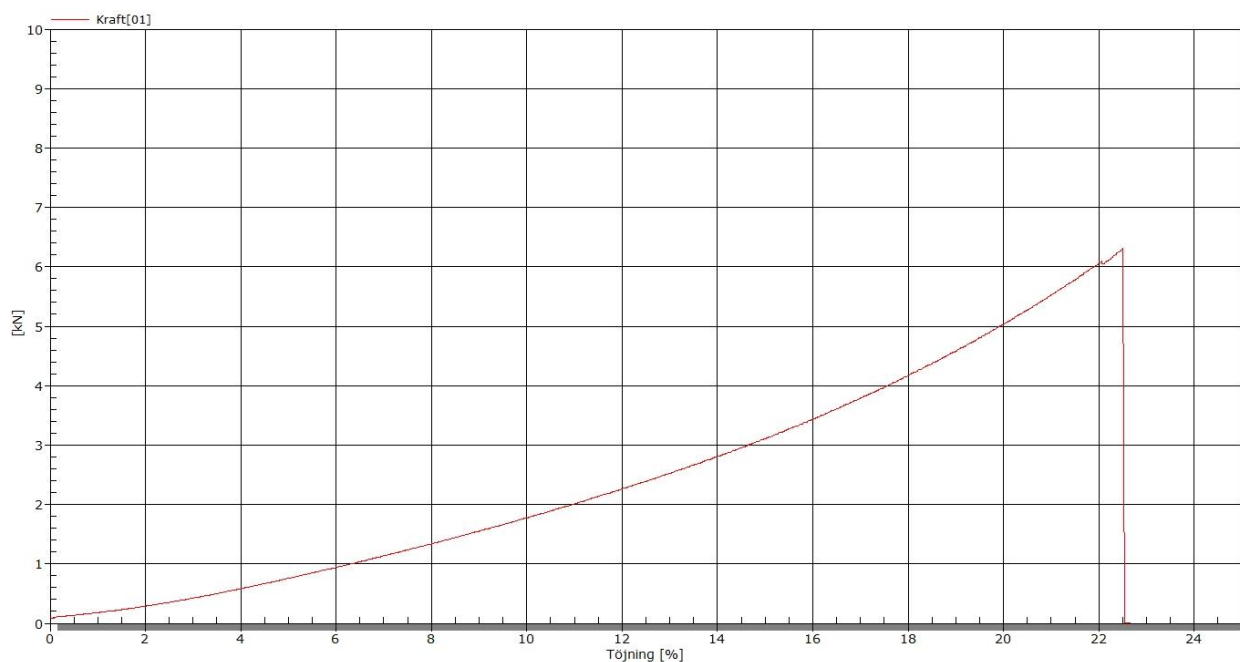


Diagram 5. Test marking No 16.

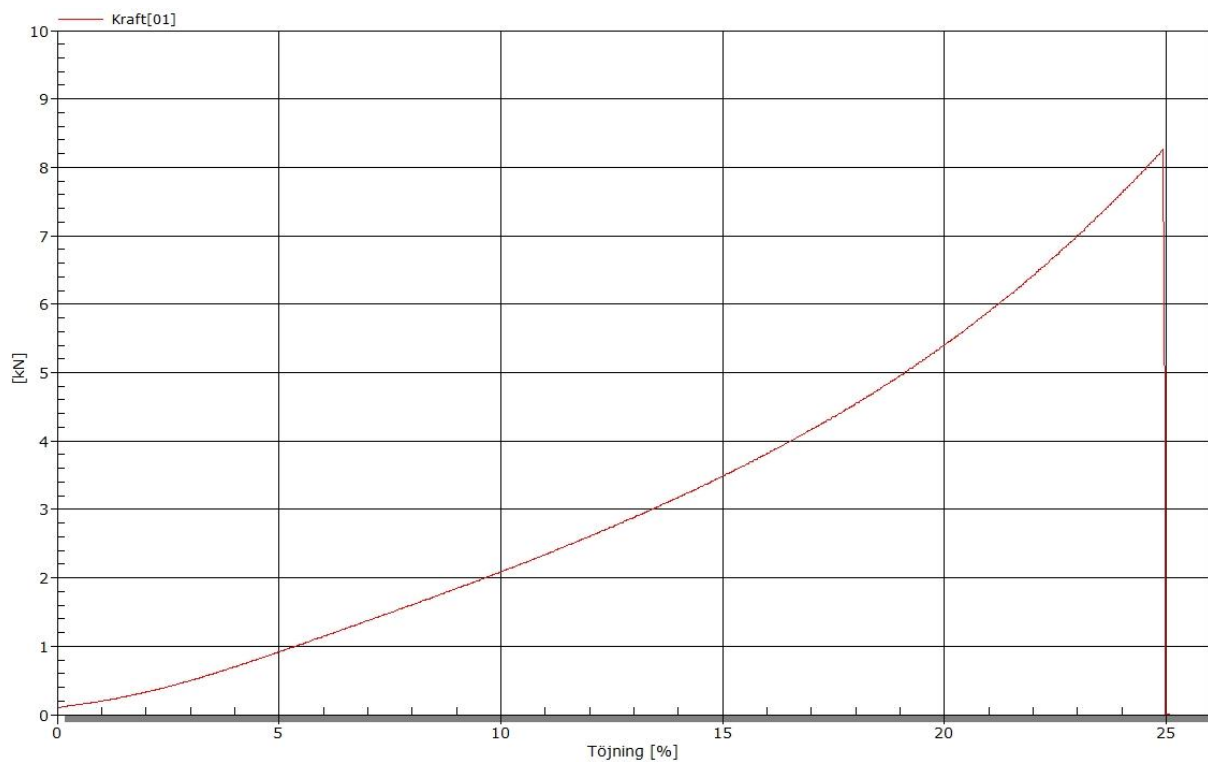


Diagram 6. Test marking No 20.